

Sign Errors in Parabola Transformations

Reading vertex form, writing transformed equations, and finding sign errors

Directions: Every problem on this sheet turns on a *sign*. Work each one carefully and, where a student's work is shown, say exactly which step went wrong before you write the correct answer. Showing the substitution you used is part of the answer.

The form everything here is written in

$$y = a(x - h)^2 + k \quad \text{vertex } (h, k), \quad \text{axis of symmetry } x = h$$

The subtraction in $(x - h)$ is part of the *form*, so the vertex x is the value that makes the parentheses zero — not the number you see printed. Decide h by asking “what makes $x - h$ equal 0?” A negative a reflects the parabola over the x -axis; it does *not* change which way the graph shifts.

- 1.** Let $f(x) = (x - 5)^2 + 2$.

First decide where the vertex is: is it at $x = 5$ or at $x = -5$? Then evaluate f at the vertex to find the **minimum value** of f . Show the substitution you used.

Answer: _____

- 2.** The graph of $y = x^2$ is translated **4 units left** and **3 units down**.

Write the equation of the new parabola in vertex form, then evaluate *your* equation at $x = -4$. What value do you get?

Answer: _____

3. Let $p(x) = (x - 1)^2 - 16$.

Find both **x -intercepts** of p . (Check your signs against the vertex: the two intercepts must sit the same distance on either side of the axis of symmetry.)

Answer: _____

4. Find and fix. Jamal is asked for the maximum value of $y = -(x - 6)^2 + 7$. He writes:

“The vertex is at $x = -6$, so the maximum is $-(-6 - 6)^2 + 7 = -137$.”

Name the sign error in his work, then give the **correct maximum value**.

Answer: _____

5. The graph of $y = x^2$ is **reflected over the x -axis**, **stretched vertically by a factor of 3**, then shifted **2 units left** and **12 units up**.

Write the equation of the new parabola, then evaluate *your* equation at $x = 0$.

Answer: _____

6. Find and fix. Mia writes:

" $g(x) = (x + 7)^2 - 5$ is $y = x^2$ shifted 7 units right and 5 units down, so its lowest point is $g(7) = 191$."

Mia made one sign error that then produced a wrong minimum. Say which shift she described incorrectly, then give the **actual minimum value** of g .

Answer: _____

7. The graph of $y = x^2$ is reflected over the x -axis, shifted **2 units right**, and shifted **25 units up**.

Write the equation, then find both x -**intercepts**. Explain in one line why the reflection does *not* move the axis of symmetry.

Answer: _____

8. Error analysis — study card. Ravi made this card for the test. Exactly one of the three rules is correct.

R1. In $y = a(x - h)^2 + k$, the graph shifts h units in the direction of the sign you see: $(x + 4)$ means right 4.

R2. k shifts the graph up when $k > 0$ and down when $k < 0$.

R3. When $a < 0$, the parabola is reflected, so all its shifts reverse direction too.

For **each** rule: say whether it is right or wrong, explain *why* in terms of what makes $(x - h)$ equal zero or what a does to the outputs, and rewrite every wrong rule so it is correct.